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ORIGINAL PAPER

# What is generative in generative artificial intelligence? A design-based perspective

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**Abstract**
Generative artificial intelligence (GenAI) models have attracted tremendous interest since the advent of ChatGPT, raising numerous opportunities and challenges. However, their generative power has not yet been studied, leaving open the question of what is truly generated by these tools. This paper addresses this question and precisely characterizes the generativity behind GenAI models. Owing to the latest advancements in engineering design, we first propose a framework for uncovering the various types of generativity. Then, we consider the main families of GenAI models and systematically analyze them to characterize their generativity within this framework. By doing so, we highlight the existence of two distinct generative levels in GenAI: one leading to the generation of new artifacts and the other leading to the generation of GenAI models themselves. We are also able to characterize the generativity of both of these levels, thus specifically confirming the generative power of GenAI and opening research avenues toward human-GenAI collaboration.

**Keywords** Generative artificial intelligence · Generativity · Design theory · Artificial intelligence

## 1 Introduction: the generative trend in artificial intelligence

Generative artificial intelligence (GenAI) models are able to "go beyond what they've seen before and create something new " said Ian Goodfellow, thus granting GenAI generativity with respect to the etymological meaning of this term: the ability to produce or create something new. Moreover, with this sentence, Goodfellow positioned GenAI in opposition to traditional AI, which was mainly considered for its ability to classify and find order in existing information. Since the advent of ChatGPT, GenAI has become the center of attention, and numerous such models that are able to generate diverse types of content, from texts to videos by way of images, are now available (Cao et al. 2023a). The so-called generativity of these GenAI tools is at the origin of many discussions, and the number of related use cases is flourishing. Among these discussions are the implications of such tools, whether social or economic, as well as technical issues such as computational times (Feuerriegel et al. 2024).

This generative characteristic of algorithms is at the origin of many hopes but also doubts, sometimes leading people to question this very trait. As a research object, generativity has been discussed for decades, and several forms have been highlighted. This research aims to address the concept of generativity in GenAI, with the objective of understanding the various forms of generativity. As such, this objective is in line with the recent work of Villani and McBurney (2024), who leveraged category theory to provide a precise analysis of transformer neural networks, which form one of the core concepts in GenAI. In our case, leaning on design theories we focus on the generativity of GenAI models. The logic being that to fully use tools (in our context GenAI ones) it is of major importance to start understanding how they work. Which is very close to Plato (1998), explaining that "no tools will […] be of any use to him who has never bestowed any attention upon them".

To achieve this objective of understanding the generativity of GenAI, the remainder of the paper is structured as follows. The second section, a literature review, looks at the existing GenAI models and introduces the concept of engineering design as relevant to characterizing their

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generativity, leading us to formulate the research question. The proposed method, which is based on a canonical model of generativity, is then explained according to an analysis of some GenAI models. The results, which precisely characterize the generativity of each tested model in terms of the canonical model of generativity, are given in the fifth section. A discussion of the literature on both artificial intelligence and engineering design then concludes the paper.

## 2 Background and scope: generativity and artificial intelligence, two disciplines that can enlighten one another

### 2.1 Generative artificial intelligence: definition and models

Artificial intelligence was built on the promise of human understanding and a construct of intelligence, whereas generative artificial intelligence (denoted as GenAI in the rest of the paper) is "the science which studies the (fully) automated construction of intelligence" (Zant et al. 2013). More pragmatically and mathematically, GenAI is a set of models designed to learn the hidden underlying structure of a dataset and to "generate new data points that plausibly could be part of the original dataset" (Pinaya et al. 2023). The definition of a model itself, both in GenAI and more generally in artificial intelligence, is not straightforward, even though authors have considered that a model is a combination of an architecture and a training phase for decades. This means that a model is obtained once its architecture and parameters are fixed, which is achieved through a training phase (Friedman et al. 2001), as is the case with famous neural networks. Following this, the training phase is part of the modeling process; as such, in the rest of the paper, we refer to a model once its training process is completed, its parameters are fixed, and it is thus able to make predictions or generate content (Ghahramani 2015).

The recent rise of ChatGPT, developed by OpenAI, opened an avenue for discussions concerning GenAI, both in the academic and public worlds. Mathematics and computer science scholars have focused on models and their technical performance, improving the existing models and developing new generative modeling frameworks (Gozalo-Brizuela and Garrido-Merchan 2023). The high promised and experienced performance of these models is leading to great hope for many use cases across various disciplines (Ooi 2023), such as medicine (Paladugu et al. 2023), supply chains (Fosso Wamba et al. 2023) and education (Baïdoo-Anu and Ansah 2023), with more general promises regarding the benefits of GenAI for helping humanity solve great challenges such as climate change (Rane 2023).

These hopes are supported by several GenAI models with outstanding performance, as they benefit from continuous improvements provided by computer science scholars. Among these models are the famous generative pretrained transformers (GPTs), especially ChatGPT and the LLaMA model developed by Meta (Yenduri et al. 2023). Another class of widely known generative models includes variational autoencoders (VAEs), which have been used for a longer period of time and in numerous content generation applications and are mainly employed in unsupervised settings (Doersch 2021). In contrast to the previously developed models based on neural networks, another widely used category of models, namely, generative diffusion models (GDMs), which are based on Markov chains, is specifically used to generate images from text with tools such as the famous MidJourney or Dall-E systems (Cao et al. 2023b). To overcome the initial difficulties faced by VAEs and to avoid the use of Markov chains, generative adversarial networks (GANs) have been introduced and are now widely used (Creswell et al. 2018). All these models, which have been developed in recent years, are said to be generative, defined as their ability to produce "previously unseen synthetic content, in any form and to support any task" (García-Peñalvo and Vázquez-Ingelmo 2023). Although this generative power is at the origin of many hopes, it has yet to be noted that some limitations have also been mentioned (Ray 2023). We should also stress that the generative power of GenAI can depend on the given tasks and the utilized model itself. This paper elucidates the generativity of such models and consequently starts to indicate their relevance for various tasks.

### 2.2 Generativity: what can engineering design bring to artificial intelligence?

Generativity and the capacity of design processes to be generative have been discussed for decades; the first so-called "generative design" dates back to the 1970s/1980s in the context of education or architecture (Gullichsen and Chang 1985; Koffman 1973), and such designs are often based on decision trees and rule-based systems.

Over the years, the engineering design community has specialized in the study of generativity, defining this concept as the transformation of known objects into new ones (Rogers et al. 2005; Hatchuel et al. 2011). Scholars have even considered the idea that generativity is the precise topic of design theories, studying the generative power of these theories (Rogers et al. 2005; Eris 2003). They have also shown that the development of formal design theories, such as the famous general design theory, axiomatic design, coupled design theory, the function-behavior-structure framework, and C-K theory, has followed a logic of enhancing generativity over the years (Hatchuel et al. 2011). The history of various design theories has been studied in terms of generativity,

and classic theories such as axiomatic design and systematic design have been categorized according to their generativity levels (Hatchuel et al. 2011). In their quest to formalize generativity, engineering design scholars have relied on the mathematics of set theory, which ultimately led them to derive conditions, namely, the splitting condition and the countable condition, for a model to be generative (Jech 2007; Masson et al. 2016).

With the expertise developed over the years by studying generativity, engineering design scholars have started to study generative algorithms. The first stream of research characterizes the generativity of algorithms (Hatchuel et al. 2021; Thomas et al. 2023), whereas the second stream evaluates the novelty and diversity of the outputs of GenAI tools (Ghasemi et al. 2023) as well as the impacts of artificial intelligence, generative and nongenerative, during the design process (Doshi and Hauser 2023; Saadi and Yang 2023; Zeng et al. 2019). However, to our knowledge, counterintuitively, no study has attempted to precisely and formally characterize the generativity of GenAI, except that of Hatchuel et al. (2021) in the case of generative design algorithms (GDAs). Hence, we wish to extend this first stream of works to other GenAI models, thereby complementing the second stream by discussing the evaluations and impacts of such GenAI models.

### 2.3 Characterizing generativity in artificial intelligence

The two previous paragraphs showed that the body of literature on GenAI is growing, with many expected use cases. However, the generative power of such models has not yet been formally characterized; hence, the question we wish to address is as follows:

*How can the generativity of GenAI models be formally characterized?*

This question is important at both the theoretical and managerial levels, theoretically because it elucidates the generative power of such algorithms and managerially because it provides insights into when and how to use each model based on the desired generativity characteristic.

## 3 Method: a canonical generativity framework for systematically analyzing GenAI models

The method we adopt to address the above question is in two steps. First, we derive a generativity framework based on the engineering design literature, which allows us to precisely characterize the generative power of each category of models. Second, we systematically analyze four categories of GenAI models by mathematically analyzing the functioning of each model.

Both of these steps allow us to formally characterize the generativity of GenAI models by first providing an understanding of what is generativity and then by understanding the mathematical foundations of each model. The remainder of this section is organized according to these two steps.

### 3.1 A theoretical framework of generativity

Generativity has long been discussed, especially by design and engineering scholars, who even consider that "the specific object of design science is generativity" (Hatchuel et al. 2011). To understand generativity, one should start with a canonical task that is at the heart and origin of design science, specifically, the design of a new artifact $A$ with some properties denoted as $P(A)$ (Hatchuel and Weil 2008). More precisely, the artifact $A$ can be understood as a set of design parameters $\left(DP_1, \ldots, DP_n\right)$ that are directly actionable by the designers and lead to a set of functional features $\left(FR_1, \ldots, FR_m\right)$ with an expected performance level; the parameters and features are both linked by a model $M$ such that $\left(FR_1, \ldots, FR_m\right) = M\left(DP_1, \ldots, DP_n\right)$. Once this design task starts, the generative process begins, for which numerous methodologies and processes have been proposed (Evbuomwan et al. 1996).

The engineering design literature provides numerous methodologies to do so, the most generic of which is C-K theory (Hatchuel et al. 2018). This theory states the existence of two disjunct spaces, namely, a concept space to be explored by the designer, which is composed of propositions without logical statuses, and a knowledge space, which contains propositions with logical statuses. One of the major contributions of this theory is the interplay between these two spaces, which has led to an understanding of the reasoning logic of designers. This theory has been used in various design tasks, both for designing products or services in industries (Hatchuel et al. 2004) and for providing a theoretical understanding of design processes and generativity (Hatchuel et al. 2011).

In the second theoretical stream, one can note that by varying the K space, C-K theory generates a variety of generativity types. When the K space is composed of a model $M$ that maps the design parameters to the functional features, it can lead to two generativity modes. The first is the activation of this model, and the second is the inversion of this model (if the model is invertible; otherwise, the inverse is estimated where appropriate). The concept of generativity by inversion has been exemplified by Redtenbacher's systematic design (Redtenbacher 1868) and it should be noted that this type of generativity is the most common since it starts with intentions (embodied by the functional requirements as starting points). Now, if the K

space is not composed of a model *M* of an object but only a language L (i.e., fixed object dimensions and a distinction between the design parameters and functional features), a new type of generativity is produced. This type is found in classic design theories such as axiomatic design (Suh 2001) and general design theory (Reich 1995). The last situation is that in which the K space is not fixed a priori: the dimensions of the artifact to design *A* can evolve, and there is no distinction between the design parameters and functional features. This is the canonical situation of C-K theory, where C and K coevolve, creating a general logic of generativity (Hatchuel and Weil 2008).

In brief, the above paragraph stresses the existence of various generativity types that can be identified from design theories. This is not surprising since "the specific object of design science is generativity", as (Hatchuel et al. 2011) highlighted. Over the past few years, the artificial intelligence community has claimed that its models can be generative; therefore, it appears natural to leverage design theories, given the understanding of generativity that they provide, to account for the generativity of GenAI. Indeed, design theories provide us with a general and theoretical framework for understanding generativity; however, we should be careful and note the limits of this approach. Since we are dependent on the available theories, we must not rule out the possibility that the use of other theories could lead to a different understanding of generativity.

Overall, this section puts forward four main types of generativity, as summarized in Fig. 1, which can be categorized into two main groups.

- The first path contains a fixed model of an artifact, within which two types can be observed. The first sub-type, generativity by activation, consists of activating the fixed model, whereas the second sub-type consists of inverting the model or at least estimating its inverse.
- The second path includes no fixed artifact model, within which two types can also be observed depending on whether a fixed language is available.

This quick review of the existing design theories reveals four main types of generativity, thus formulating a framework that allows us to understand and characterize the generativity of GenAI algorithms.

## 3.2 Some illustrations of this framework

In this short subsection, we illustrate the proposed framework in action in the context of a generative task. To become more familiar with this framework, we therefore propose three kinds of illustrations: first, canonical illustrations that introduce standard situations for each of the types; second, an illustration based on a common starting design task to highlight the differences between the types; and third, an illustration in a more computer science-related setting, focusing on generative design algorithms (Hatchuel et al. 2021; Thomas et al. 2023).

### 3.2.1 Canonical illustrations of each type of generativity

Let us start with canonical examples of each of the types of generativity. These illustrations are mainly inspired by the literature, especially from historical situations described in the design literature.

The first type of generativity, namely, *design by activation,* is very similar to drawing according to, for instance, a Gaussian law. This can be understood since a Gaussian law is a known model with a specific probability density function, which can be obtained from Monte Carlo simulations based on uniform laws over the range of [0,1] (Harrison 2010). Here, one can understand that the model is known (with a Gaussian probability density function, uniform laws, and a transfer function from uniform laws to a Gaussian law), and the generativity type consists of activating it.

The second type, namely, *design by inversion*, can be effectively illustrated by Redtenbacher's systematic design in the context of hydraulic wheels (Redtenbacher 1868). This German engineer used charts (he had previously designed them himself) to respond to customers' wishes for hydraulic wheels. In other words, as made very clear by Masson

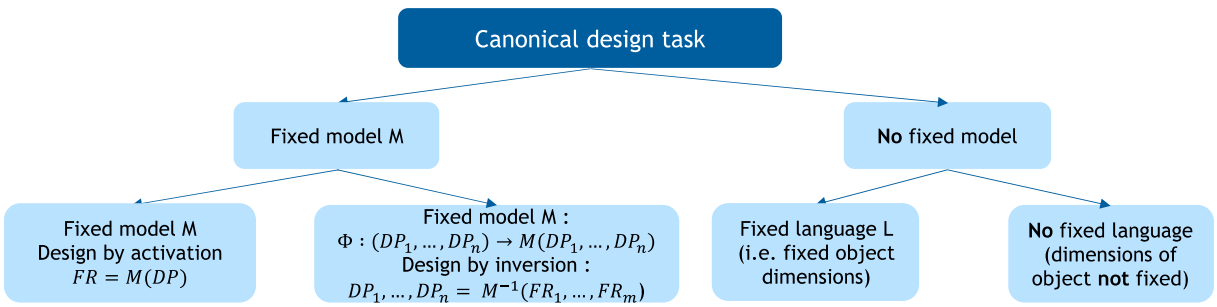



**Fig. 1** Framework of the various types of generativity considered

and Weil (2013), Redtenbacher first created a model of the "hydraulic wheel" object and then started by asking a set of questions to customers. The responses to these questions allowed him to set the functional requirements of the wheels (for instance, the price and river current), enabling him to select a wheel class. In other words, once the wheel model was designed, he started from the functional requirements, allowing him to select the best design parameters for achieving the expected performance.

The third type, namely, *fixed-language design*, can be well illustrated by Suh's axiomatic design (Suh 2001). In this famous design task, the author proposed a matrix-based approach to map design parameters and functional requirements. One canonical example is the design of a mixer tape. The functional requirements (flow and temperature) and the constraints (hot and cold-water sources) are known. A classic two-faucet solution can be implemented, but axiomatic design would indicate the design of a new working model based on a diagonal matrix, leading to the desired mixer tape. Therefore, here, a generativity type takes a language of design parameters and functional requirements as inputs, and the best model is designed to map one group to the other.

The fourth type of generativity, namely, *nonfixed-language design*, is a type in which the previous language of design parameters and functional requirements is not available or at least can be revised. This is typically what is suggested by C-K theory (Hatchuel and Weil 2008), as in the case of an engine used on Mars, which led to the design of a Mars hopper, an artifact for which a language was not even initially known.

### 3.2.2 Illustrations of each generativity type with a common design scheme

In addition to these four canonical illustrations, we propose an illustration starting from the same design task to illustrate the discrepancies between these generativity types. This is inspired by the analysis performed by Hatchuel et al. (2011), where the authors compared the generativity levels of various design theories and illustrated them in a situation involving a lighter and less expensive camping chair.

- The first type consists of activating the model of a chair, thus linking the design parameters to the functional requirements and indicating the use of lighter and less expensive materials. Typically, this would lead to the design of a chair with less materials used, for instance. This appears to be very close to the work done by Barros et al. (2014), where the authors were able to generate millions of chairs by starting with a parametric model of a chair (with nine design parameters concerning the backrest (angle, width…), set (height, diameter…) and legs (angle)). Yet, it is to be noted that among these millions of chairs, the designer needs to select one.
- The second type, on the other hand, involves starting from the functional requirements, for instance, "a chair weighing less than 1 kg". Inverting the known model would, for instance, indicate the use of an all-aluminum chair, as this material is known for being both light and inexpensive. One distinction to be noted with the first type is that having a constraint on the functional requirement spares selection, unlike in the type of design by activation.
- The third type typically resorts to an axiomatic design framework, with the objective of decoupling a matrix. For instance, a canonical situation would be to add a new function to this "lighter and less expensive campaign chair", such as "can be dismantled". Such a new function would couple the matrix and require the design of new conceptual chair models while remaining in a fixed-language framework.
- Finally, the fourth type of generativity is based on a new discussion of a commonly accepted characteristic of chairs. For instance, the designer could look for "a chair that makes sitting on the floor quite comfortable" (which is unusual since one should not sit on the floor with a chair). This would lead to rediscussing the very concept of a chair, therefore leading to the proposal of a new definition of what a chair is and the concept of a "chairless chair", as illustrated on the right side of Fig. 2. The main distinction between this type of generativity and the previous one of *fixed language* is that the latter does not question the dominant design. In the example of the camping chair, the *fixed language*

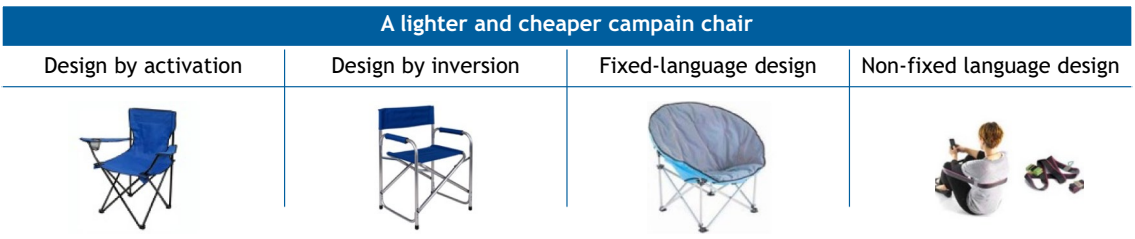



**Fig. 2** Illustrations of the various designs obtained from different generativity types for a "lighter and less expensive camping chair"

*design* type would not allow to reach a chair without legs, whereas this is achievable in the *nonfixed language design*.

#### 3.2.3 Illustration of the whole framework with GDAs

To illustrate how the proposed framework can be useful for classifying the generativity of algorithms, we propose using GDAs. We refer to the work of Hatchuel et al. (2021); Thomas et al. 2023), published at recent ICED conferences, that formally characterized the generativity of various algorithms. Specifically, they showed that GDAs, including MapElite and nondominated sorting genetic algorithm (NSGA)-II, tend to generate artifact topologies. Such an artifact topology is generated by inverting a fixed model, and as such, the model enters the second path illustrated in Fig. 1: *design by inversion*. The authors explained that the model considered in a GDA is often noninvertible, which, as we have seen above, leads to the creation of an estimate of its inverse.

### 3.3 A systematic analysis of GenAI models

To understand the generativity hidden behind GenAI models, we resort to a systematic analysis, for which three main points must be remembered. First, we focus on families of models instead of one model, where a family of models is defined as a set of models sharing a common philosophy and architecture, as we have seen in the literature. Second, we concentrate on four main families of models, namely, VAEs, GDMs, GANs and GPTs. We choose these families because they are the most widely used models in practical applications as well as the model families that are most commonly studied by academia, and they handle a wide variety of data (text, images, sounds, etc.) (Cao et al. 2023a). Third, to systematically analyze each family, we concentrate on the mathematical foundation of the seminal model of the family, explicating its architecture, training phase and associated generative procedure. To do so, because we focus on families of models instead of one specific model, we base our analyses on review papers.

The following section is dedicated to this analysis, and to provide clarity, we adopt a systematic organization scheme: each subsection is dedicated to one family of GenAI models, within which the first paragraph is dedicated to a mathematical analysis. In the following section, we denote the input data by $X$, the output data of the model by $Y$ and the considered GenAI model by $M$. The overall philosophy of GenAI models is to learn the model $M$ based on the input data $X$ to generate new data $Y$.

## 4 Artificial intelligence models within this framework

### 4.1 Objective and scope of the section

As stated above, the objective of this section is to precisely describe the mathematical foundations behind each of the retained families of GenAI models, which will allow us to formally characterize their generativity and position them in the framework summarized in Fig. 1.

### 4.2 Generativity of VAEs

#### 4.2.1 Mathematical foundation

VAEs were models introduced in 2013 (Kingma and Welling 2019) and are now widely used to generate content, especially images. The following papers form the basis of the analysis provided in the subsequent paragraph: (Yenduri et al. 2023) and (Barros et al. 2014).

The overall process of a VAE is to represent a real data distribution $P_X$ in a smaller space $\mathcal{Z}$, which is a latent space where the distribution $P_X$ is approximated by $P_Z$. To do so, a VAE is composed of two parts: a neural network encoder whose objective is to learn to embed $X$ into $\mathcal{Z}$ and a neural network decoder whose objective is to reconstruct the original distribution $P_X$ from $P_Z$ (Kingma and Welling 2019). The distribution $P_Z$ is often close to a Gaussian-centered distribution since in dimension d, it is known that all distributions can be obtained from a set of Gaussian-centered distributions and a (possibly complicated) function.

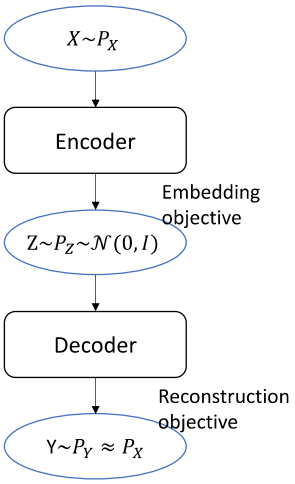
The generative procedure described in Table 1 provides a schematic overview of the method used to construct such models.

#### 4.2.2 Example and characterization of generativity

Let us start with a simple example, as given in Doersch (2021), involving the well-known MNIST database consisting of handwritten digits. A VAE, as described in Sect. 4.2.1, was trained on this database using the sigmoid cross-entropy loss as the loss function. This led to the convergence of the model parameters after completing the training phase. Finally, by feeding Gaussian noise to the model, a sample of generated handwritten digits could be obtained.

Overall, in a VAE, a model is learned through a dual encoder–decoder structure after executing a training phase to represent the underlying distribution of the initial

**Table 1** Generative procedure of VAEs

| VAE: generative procedure | |
|---|---|
| *Input:* A set of data $X$ with an initial distribution $P_X$<br>*Architecture:* An encoder $E$ and a decoder $D$, the first of which embeds $X$ with a complex distribution $P_X$ into $Z$ with a simple distribution $P_Z$(often $\mathcal{N}(0, I)$)<br>*Training phase:* Both $E$ and $D$ are trained simultaneously with the objective of minimizing the deviation between $P_X$ and $P_Z$ (measured by the Kullback−Leibler divergence measure) for the encoder for $E$ and the reconstruction loss for $D$. Overall, the objective is to minimize the difference between $X$ and its final reconstruction $Y$.<br>*Generation phase:* One data point is sampled following $P_Z$ (often $\mathcal{N}(0, I)$) and mapped to $P_X$ through $D$, thus designing a new artifact $Y$ |  |

training database (in the above example, the underlying distribution of the MNIST database is learned). Then, in a strict sense, generativity comes from the decoder, which is activated when feeding noise to the model. As such, this family of models enters the first path of Fig. 1: *design by activation.*

## 4.3 Generativity of GDMs

### 4.3.1 Mathematical foundation

GDMs were initially introduced in 2015 (Sohl-Dickstein et al. 2015) and are now widely used for generating images from a sentence, as in the famous text-to-image generator Dall-E. The following papers form the basis of the analysis provided in the subsequent paragraph: (Doersch 2021; Sohl-Dickstein et al. 2015; Chang et al. 2023), and (Yang et al. 2023).

The overall idea behind such models is to add noise to an image based on probabilistic diffusion models; this procedure is inspired by thermodynamics and includes three components: a forward process, a reverse process and a sampling procedure. The forward process gradually adds noise to an image $X$ over $T$ time steps due to a transition kernel until a Gaussian distribution $\mathcal{N}(0, I)$ is produced for $X_T$. The reverse process operates based on the inverse of the transition kernel to denoise the image obtained at time $T$, and the parameters of this kernel are learned by neural networks. The generation strategy is based first on sampling from $X_T$ and then denoising it according to the reverse chain. Notably, unlike VAEs, which attempt to embed the original distribution into a smaller latent space, GDMs are not based on dimensionality reduction but rather attempt to estimate a space (through noise) that encapsulates all image variations. This mechanism can be improved by adding text guidance, as done by text-to-image generation tools such as Dall-E, in which the input text is understood by a transformer; this method is developed hereunder. Several techniques exist, but the main strategy is to train a classifier that assigns weights to constrain the reverse denoising process. A classifier-free method that is more computationally friendly has also been derived and performs conditional extrapolation according to the labels of the given input sequence.

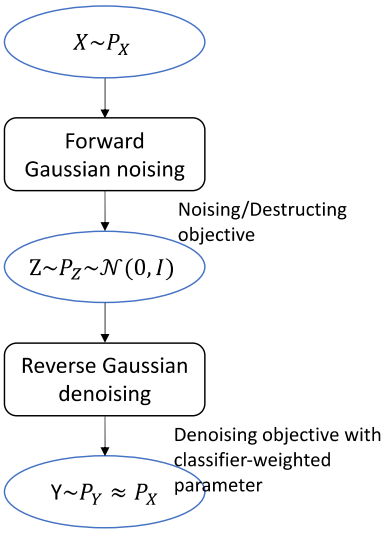
The generative procedure is described in Table 2, along with a schematic overview of the working method of such models.

### 4.3.2 Example and characterization of generativity

Let us use the example of "a corgi wearing a red bowtie and a purple party hat" from Nichol et al. (2022). To obtain this surprising image, the authors provided this input text sequence to their previously trained text condition-based diffusion model. Their model was trained on the same input database as that used for the famous DALL-E tool, and they employed an ImageNet network to optimize and learn the parameters of the Gaussian reverse process.

Overall, in GDMs, an initial noising step is carried out to transform the initial distribution into a Gaussian distribution. The parameters of the reverse Gaussian distribution are trained in GDMs, providing a fine model of the artifact to be designed. This second step is sometimes a text-guided process, for instance, a classifier weighting its parameters, as is the case in the example given in the above paragraph. In this example, it appears that the training phase was aimed at learning a model of the to-be-designed artifact, in that case

**Table 2** Generative procedure of GDMs

| GDM: generative procedure | |
|---|---|
| *Input:* A training set of data $X$ with a distribution $P_X$, $X_0 \sim X$ and an input text sentence<br>*Architecture:* Two main processes: a forward noising step transforming $X_0$ into $X_T \sim \mathcal{N}(0, I)$ by progressively adding Gaussian noise and a reverse denoising process learned to reduce noise to return to $X_0$. The denoising process is possible due to the Gaussian distribution, whose parameters are learned to minimize the discrepancy between the reconstructed data and the initial data. In this second step, several text condition-based techniques are applied by weighting the added noise with a classifier<br>*Training phase:* The denoising process needs to be trained on a set of data to find the optimal parameters<br>*Generation phase:* One data point is sampled from the final distribution of the noise addition phase (often $\mathcal{N}(0, I)$), and then the reverse denoising step is applied to return to $P_X$. |  |

a corgi, from a set of images. In conclusion, generativity is achieved through a denoising process, and as such, such models can enter the type of *design by activation.*

## 4.4 Generativity of GANs

### 4.4.1 Mathematical foundation

GANs were introduced in 2014 by Goodfellow et al. (2014) and are currently the most widely used GenAI models for generating images, sounds or videos. The following papers form the basis of our analysis: (Nichol et al. 2022; Goodfellow et al. 2014; Arora et al. 2018; Gui et al. 2023).

The overall way in which these models work is to learn a probability distribution $P_X$ that generates the given set of training data $X$. Although the seminal papers stated that GANs learn such a distribution, it has been precisely shown that they only learn almost all of the distribution (see Arora et al. 2018, 2017) for precise mathematical details on what is learned by GANs). The probability distribution associated with $P_X$ (which is formally characterized here by its probability density function), which is often complex, explains why classic density estimation techniques, such as maximum likelihood estimation, are challenging in practice. GANs build on the framework of game theory, where the two players are neural networks, thus forming the two parts of a GAN: a neural network generator $G$ that is trained to map noise (following $\mathcal{N}(0, I)$) to $P_X^G$, which is a close estimate of $P_X$; and a neural network discriminator $D$ that is trained to distinguish between $P_X$ and $P_X^G$. The key point to keep in mind is that GANs learn to generate data as if they were drawn from the true data distribution. The subtlety of GANs resides in the training phase since both $D$ and $G$ are trained simultaneously; $D$ is trained to maximize its classification accuracy, in other words, to correctly distinguish between real samples and fake samples produced by the generator, and $G$ is trained to minimize the classification error, i.e., to produce samples that will be classified as real by the discriminator. As such, both neural networks are co-trained, and the model continues to evolve with the generated samples.

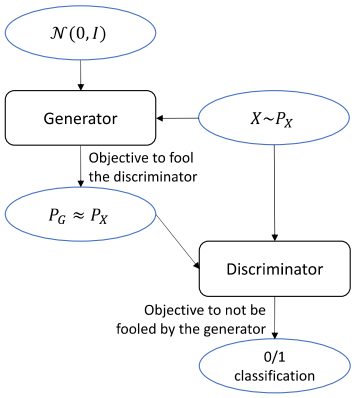
The generative procedure is described in Table 3, along with a schematic overview of the working method of such models.

### 4.4.2 Example and characterization of generativity

Let us again start with a canonical example from the MNIST database, as described by Goodfellow et al. (2014). The authors started by designing two neural networks, a discriminator and a generator, and trained them on this database until they converged to optimality. In other words, until convergence, the probability distribution learned by the generator was sufficiently close to the real distribution. Once this state was obtained and both networks were optimized, they fed the noise to the generator, which allowed it to generate a new image of a handwritten digit.

Overall, in a GAN, a model of the to-be-designed artifact is learned by the generator due to its co-training process with the discriminator. This co-training strategy is based on the ideas of game theory, with the generator and the discriminator as the players. The training phase ends when the optimum is reached; this optimum allows the generator to learn a probability distribution that is shown to be close to the real probability distribution of the initial database. Afterward,

**Table 3** Generative procedure for GANs

| GAN: Generative procedure | |
|---|---|
| *Input:* Real dataset $X$ with a probability distribution $P_X$<br>*Architecture:* A neural network discriminator $D$ trying to differentiate between the distributions $P_X$ and $P_X^G$ and a neural network generator $G$ whose objective is to learn an estimate $P_X^G$ that is as close as possible to $P_X$<br>*Training phase:* Both $D$ and $G$ are trained simultaneously to maximize the probability that a sample is real and minimize the probability that $D$ estimates the sample to be real<br>*Generation phase:* With the learned distribution, the generation phase can be implemented by sampling by the generator $G$ |  |

the generation step is carried out by feeding the generator an input (generally noise) and sampling according to the estimated and optimal distribution. Consequently, this learned distribution represents a model of the initial database, which allows GANs to perform *design by activation.*

## 4.5 Generativity of GPTs and large language models (LLMs)

### 4.5.1 Mathematical foundation

GPTs and LLMs form the most recently developed family of GenAI models; they were introduced in 2018 by the OpenAI company (Radford and Narasimhan 2018), who built on the work of Vaswani et al. (2017) and is at the origin of the current interest in GenAI, especially with regard to text understanding and generation. The following papers form the basis of our analysis: (Rane 2023; García-Peñalvo and Vázquez-Ingelmo 2023; Arora et al. 2017; Vaswani et al. 2017; Brown et al. 2020; Dehghani et al. 2019), and (Lin et al. 2022).

The overall architecture of the original transformer, as initially defined by Vaswani, et al. (2017), is composed of an encoder $E$ and a decoder $D$, along with an important attention mechanism aimed at understanding the word sequences in input text data. As such, the architecture is similar, yet far from identical, to that of a VAE since a GPT relies on a crucial attention mechanism to generate meaningful sequences of words. This mechanism allows the dual encoder-decoder structure to focus on specific sections of the input by dynamically assigning them weights. Consequently, the encoder $E$ creates hidden states composed of data derived from the previous hidden state and the current input token. Then, the attention mechanism computes the relevance of each hidden state of the encoder, finally creating a context vector. This new vector is then given to the decoder $D$, which produces a token for the output sequence. This whole process is performed recursively until a whole sequence of tokens is generated. Training such an architecture only on labeled data in a supervised framework would be very costly, which is why GPTs are trained through self-supervised learning on large sets of unlabeled data. That is, they learn by predicting the next word in a sequence based on the previous words. Moreover, GPTs leverage the transfer learning framework (Zhuang et al. 2021), which consists of an initial unsupervised learning phase followed by a domain-specific supervised learning phase.

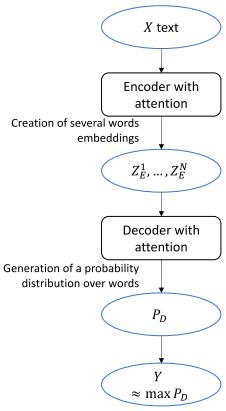
The generative procedure is described in Table 4, along with a schematic overview of the working method of such models.

### 4.5.2 Example and characterization of generativity

Let us start with an example from the seminal paper (Vaswani et al. 2017) in the context of English–French translation. The authors started by defining their encoder–decoder architecture with the attention mechanism described above. The training phase was performed by relying on a database consisting of 36 million sentences and split into 32,000 tokens (that is, words). This training phase therefore aimed at learning a model for mapping English to French; consequently, when the translation process from English to French was completed, the model could operate by activating what it had learned.

Overall, GPT models are trained and finally learn to estimate a probability distribution over a sequence of words. This learned probability distribution effectively acts as a model of human language and is able to understand the context of what has been asked in the input text. Therefore, the generation procedure itself consists of sampling according to this probability distribution, i.e., a *design by activation* generativity.

**Table 4** Generative procedure for GPTs

| GPT: generative procedure | |
|---|---|
| *Input:* Text input $X$<br>*Architecture:* An encoder $E$ and a decoder $D$; the former creates several word embeddings, which are weighted according to an attention mechanism, and the later generates a probability distribution over the relevant words in the given sequence<br>*Training phase:* Self-supervised training is first performed, and this is followed by a task-specific and supervised fine-tuning training process<br>*Generation phase:* Autoregressive generation is executed based on the architecture of the decoder $D$, and the next token is generated according to the learned probability distribution of the possible tokens |  |

## 5 Results: a generic framework for uncovering various generativity levels in GenAI

After the analysis conducted in the previous section, this fifth section is aimed at providing an answer to the initial question: what is generative in GenAI. We distinguish two main results of this work. First, two different levels of generativity exist in GenAI, both of which we are able to formally characterize according to the framework shown in Fig. 1; second, this framework is generic in the sense that it can be applied beyond the four families of GenAI focused on herein. The following subsections are dedicated to more precise explanations of these results.

### 5.1 Two levels of generativity in GenAI

The analysis above shows that there are two levels of generativity in GenAI. The first level, which we characterized, consists of the generativity of the learned model, whereas the second level concerns the model design phase. These two different levels of generativity are summarized in Table 5, and we intend to describe them more precisely in the following paragraphs.

#### 5.1.1 First level: the generativity of the learned model

First, we highlight the generativity of the GenAI models in each family, which we have been able to position in the framework of generativity, as summarized in blue in Fig. 3. All these GenAI models employ a type of generativity categorized as *design by activation*. As we have seen, this can be understood by the fact that all the families of GenAI models studied here sample over an estimated distribution, which is supposed to represent the distribution of the training database. The discrepancy between the estimated distribution and the real data distribution is what leads to the generation of previously unseen artifacts but remains as such only in the domain of existing knowledge. This level of generativity is accessible to everyone and leads to the creation of new artifacts such as images or texts. It is the level at the origin of the "wow effect" one can experience when using the now classic tools such as ChatGPT or Dall-E.

#### 5.1.2 Second level: the generativity of the model design phase

However, behind this first type, this work elucidates the existence of a second and more surprising level of generativity. This is the generativity of the so-called model design phase, which leads to the generation of a new model that will be activated to generate new artifacts (such as images or texts). This second level of generativity is

**Table 5** - Summary of the two levels of generativity in GenAI

| | Generativity of the learned model | Generativity of the model design phase |
|---|---|---|
| What is designed? | A new artifact (image, text…) is generated | A model is generated |
| How is it accessible? | Accessible to everyone, with generativity characterizing the model itself | Accessible at the convergence point of the training phase |

different from the first level and shows a *fixed-language* generativity (see yellow in Fig. 3), leading to the existence of new and previously inaccessible models. This second level of generativity enters the third type given in Fig. 1 because all the models described in Sect. 7 are created after a thorough optimization process occurs during the training phase. This training phase is based on the defined architecture of the constructed model that has the objective of converging toward the best possible model for representing the input database on which it is trained. This is how it leads to the definition of impressive models, unlike any that existed before, that are able to represent human language for instance, such as ChatGPT. This second level of generativity appears hidden by the impressive and widely accessible results obtained by the first level but is amazing in terms of its capacity to combine numerous pockets of knowledge (from statistical physics in GDMs to basic linguistics in the attention mechanisms of LLMs) to define brand new models.

## 5.2 A generic generativity framework for understanding GenAI

Another result of this work lies more in the framework itself in the sense that it appears to be generically applicable to understand the generative logic involved in the current context of GenAI, thus also allowing us to address the question of generativity versus novelty. We briefly illustrate the generality of the framework summarized in Fig. 1 with several recent papers. First, we present some models that were very recently designed to specifically address certain issues; then, a longer discussion is provided based on a paper that proposed a novel weather prediction approach. We then look at one of the most recent (up to today) models for generating video from text before finally examining an adversarial example generation case.

### 5.2.1 Generativity for ad hoc problems: solving geometric problems and protein families

The first paper we concentrate on proposed a model for generating mathematical solutions to geometrical problems given in the mathematical Olympiad (Trinh et al. 2024). This model was based on a GPT/LLM approach with a set of geometric rules. A transformer language model was used to output the final proof of the given exercise and is based on the auxiliary construction framework, which is a long-standing approach in geometry. The idea of the overall model is to generate synthetic data based on a symbolic deduction engine that generates all possible subsequent statements given the premises of a theorem that are fed to the LLM and so on until reaching the conclusion of the theorem (or a maximum number of iterations). Overall, the model relies on both a transformer and a set of given algebraic and geometric rules, thus showing a type of generativity of *design by activation*.

The second paper we propose to look at aimed at generating protein designs, which is a rather common task in GenAI (Truong and Bepler 2023). The model proposed in this paper is based on a transformer architecture but has a double attention mechanism: one for intersequence token positioning and one for intrasequence positioning. Overall, the model was trained on a set of 29 million protein sequences and learned the probability "distribution over protein families, where

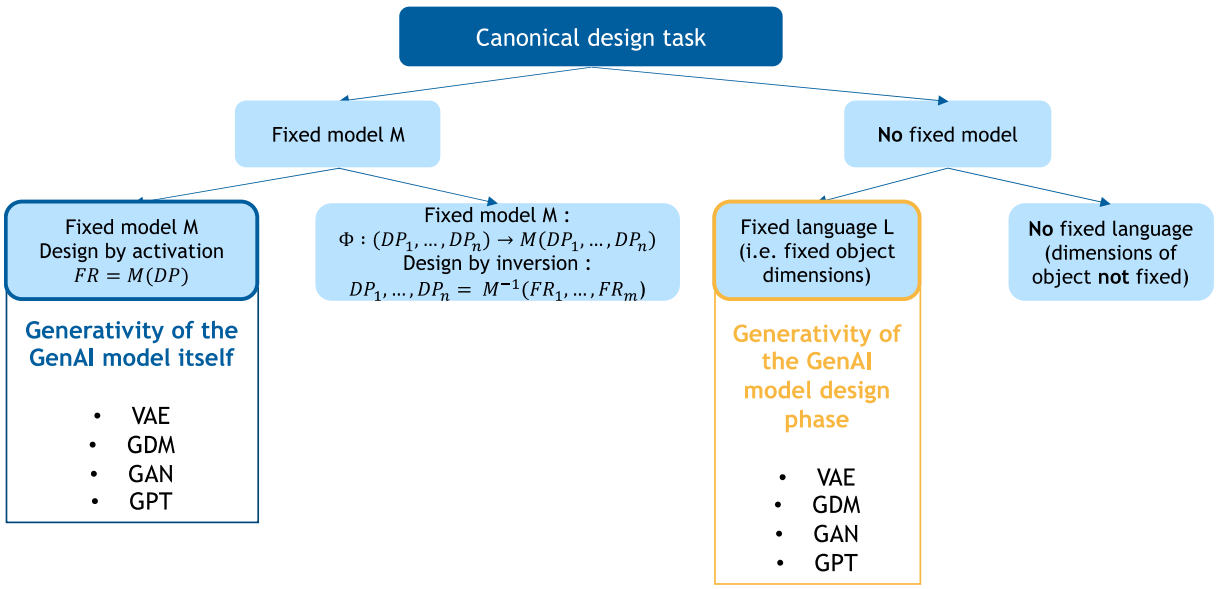



**Fig. 3** Summary of the characteristics of the two levels of generativity in GenAI: the model itself and the model design phase

each family is generated as a sequence-of-sequence". Once this probability distribution was learned, the authors generated 1000 proteins by sampling according to this distribution, what follows a generativity of *design by activation*.

#### 5.2.2 Generativity for weather forecasting

The second paper on weather forecasting represents a major paradigm shift in the world of global weather prediction in the sense that it uniquely proposed a model based on neural networks with an encoder-decoder architecture, whereas more traditional models rely on various physical equations (based on the Navier−Stokes, vorticity and thermodynamics equations) (Bauer et al. 2015; Bounoua Lahouari 1995). Consequently, a new model for inputting a weather state with classic dimensions (including temperatures and wind components) was proposed, and this model was expected to predict a future weather state with precisely the same dimensions—temperature and wind components. As such, this modeling logic appears to enter the generativity of *fixed-language design* and is close to the second level of generativity proposed for GenAI models. However, one must be careful, as this generativity is the one of the model design phases (see Sect. 5.1.2 and the yellow box in Fig. 3) and does not precisely characterize generativity itself. The architecture of this model is based on an encoder-processor-decoder architecture used to successively predict several states of weather based on the previous predicted state and the initial input weather state. This was achieved by meshing the surface of the globe to represent the relationships between the weather at different points on the terrestrial surface. The model was then trained on 39 years of weather predictions using the famous gradient descent algorithm. In conclusion, the model was trained to learn the evolution trends of weather and the links between the weather at various points on the globe, accounting for the evolution of weather over time. As such, this model once again demonstrates the generativity of *design by activation*.

#### 5.2.3 Generativity for text-to-video generation

For this last illustration of our framework, we focus on one of the most recent text-to-image generation models, the Lumiere model, which was released in February 2024 (Bar-Tal et al. 2024). Very broadly speaking, as described in this paper, this model is based on the diffusion model principle combined with a U-Net-inspired architecture. The idea is that this neural network architecture embeds video into "spacetime visual patches" that enable the efficient representation of a whole idea. As transformers work on tokens, representing sequences of words, such a model works on these "spacetime visual patches", a sequence of such patches representing a video. Then, the model relies on the principles of diffusion models. This Lumiere model was given a set of input noisy spacetime patches and a text prompt. Then, the diffusion model was trained to successively denoise these visual patches through a process guided by the prompt text. The model was trained on an impressive database of 30 million videos, all with their associated text captions. In conclusion, the denoising process was guided and trained, which was then activated by a set of noisy spacetime visual patches alongside the prompt text. Thus, this model again follows the generativity type of *design by activation*.

#### 5.2.4 Generativity in adversarial example generation

Let us end this subsection with the current developing trend concerning adversarial examples, a framework that enters the generativity of *design by inversion*, unlike all the models we have seen in Sect. 4. Indeed, we recall that our results show that among the studied families of GenAI, all of them enter the *design by activation* and none of them enter the *design by inversion* generativity. However, it should be noted that some initiatives can be found in this latter type of generativity, for instance, the works of Geng and Veerapaneni (2022); Goodfellow et al. 2014; Szegedy et al. 1312). The first paper (Geng and Veerapaneni 2022) proposed a strategy for building an adversarial example that can trick a trained network when given the expected output. This first method led to the creation of an image that looks like noise to humans, which the authors refined to create adversarial examples that were more meaningful to humans; that is, an image of an "8" is recognized as a "2" by the given network. This was also the logic adopted by Szegedy, et al. (2015) when stating that a given neural network, can be tricked by adding slight perturbations to the input images, which will therefore be misrecognized by the network. More recent developments have been made regarding the generation of adversarial examples; these works proposed generative models, inspired by GANs, that are able to systematically build adversarial examples. As such, these models are able to invert any learned classifier (Xiao et al. 2019). This last model, called AdvGAN by the authors, consequently enters the generativity of *design by inversion*, unlike previous families of GenAI models studied in Sect. 4. Even more recent works, for instance by Samvelyan, et al. (2024), combined evolutionary algorithms (whose generativity has already been characterized by Hatchuel et al. (2021) and mentioned in Sect. 3.2.3) and neural networks to generate adversarial examples given an input LLM; this method once again enters the *design by inversion* generativity type.

We conclude this illustrative subsection by recalling that the objective was to illustrate the generality of the framework introduced in Fig. 1. We analyzed four very recent papers from different streams of research that were designed

for specific purposes (specifically, geometric, protein design and weather prediction problems), a more general paper that represents one of the latest advancements in GenAI to date (specifically, text-to-video generation), and the current research on adversarial example generation. We have shown that the introduced framework allows us to effectively understand the generativity of all these models, thus demonstrating its relevance to the study of a wide range of generative models. However, one should be careful, and novel developments in GenAI in the coming decades (or years) may unveil new types of generativity.

## 6 Discussion

This last section before the conclusion provides a discussion concerning the literature, as we believe this work has many implications. First, it allows us to discuss the limits of GenAI, which relies on theoretical foundations. Second, it allows us to discuss some implications regarding human-algorithm collaboration in generative endeavors. Third, it reveals a major modeling shift that we intend to characterize.

### 6.1 Confirming the generative power of GenAI models and their limits

In this paper, based on design theory, we uncover and classify the different generative mechanisms associated with the most commonly used GenAI tools. Even if there was little doubt, our first contribution is to confirm that GenAI models are generative tools in the sense of design theory; namely, these tools can "transform known objects into new ones" (Rogers et al. 2005). In other terms, starting from a known dataset (whether made of images or texts), GenAI models are able to create new ones, this is what we have been able to observe throughout the examples given in 4.2.2, 4.3.2, 4.4.2 and 4.5.2. This underlines the notion that these tools may be particularly useful for performing design tasks.

Our results also highlight that the generative mechanisms of GenAI tools are very similar to one another tool and therefore do not cover the variety of generative mechanisms involved in design science. This contributes to the ongoing debate concerning GenAI capabilities: if GenAI tools clearly differ based on the types of objects they generate (texts, images, product designs, etc.), they do not truly differ in terms of their generative mechanisms, thus responding to the different calls for a better understanding of the capabilities of GenAI.

We also show that all generative mechanisms embedded in GenAI tools rely on a fixed model $M$ of the artifact to be designed. GenAI activates the model $M$, whereas a GDA locally inverts it. However, design theory indicates that, to achieve breakthroughs, the generative process should rely on a model $M$ of an artifact that is not fixed, suggesting that different GenAI tools using fixed artifact models have limited generative power. At a general level, this confirms some limits already identified in the literature, such as the hallucinations of ChatGPT (Wu et al. 2023). Moreover, it brings into question the combinatorial generativity of these GenAI models, which is defined as the ability to combine existing known bricks (such as distant physical knowledge) to create new artifacts (Carvajal Pérez et al. 2018). For example, GenAI models alone cannot create new physical laws based on existing laws. However, the second level of generativity highlighted above, which is more of an exercise in human design with a *fixed language*, presents such combinatorial generativity in the sense that model designers are able to incorporate physical knowledge and constraints into their generative models (Yang and Perdikaris 2018).

Nevertheless, we acknowledge that some authors have demonstrated the existence of combinatorial generativity through experiments conducted with GPT models (models at the origin of ChatGPT) regarding the generation of early-stage design concepts (Zhu and Luo 2023). In this paper, journal version of the paper (Zhu and Luo 2022) presented in 2022, before the advent of ChatGPT, the authors built on the creative cognition literature and proposed forcing models like GPT2 and GPT3 to resort to distant knowledge or an analogy-driven design to generate new design concepts. This logic can also be found in Zhu et al. (2023), where the logic of bioinspired design was implemented based on fine-tuning logic in the language model (GPT-3 in their experimental design) to generate more original concepts. Notably, both of these papers relied on fine-tuning logic and thus involved humans in the loop, which highlights the importance of human-GenAI collaboration for fostering generative endeavors, as highlighted by Zhu and Luo (2023).

### 6.2 Interactions between humans and GenAI for fostering generativity

By classifying the generativity type associated with each GenAI tool, our research suggests some room for improvement.

Effectively, our research has not yet allowed us to identify GenAI tools with a type of generativity of *design with no fixed language*, while design theory indicates that opening the dimensions of the artifact to be designed is a path toward breakthrough innovations. Moreover, design theory is a rich body of literature that, for each structure of knowledge (activation, inversion, fixed object language, or nonfixed object language), identifies some guidelines for improving generativity, from which initial insights can be drawn. As such, the obtained characterization of generativity appears to be a first step toward deriving insights regarding the use of such algorithmic tools by humans. The first generativity

type of *design by activation* opens the way to considering human-algorithm interactions, especially the increasing role of human sensemaking (Verganti et al. 2020) in this context. One of the key points, which dates back to the 1990s (Hatchuel and Weil 2011), is that the introduction of such new technologies must not be decoupled from the organization structure. In particular, the authors highlight the importance of humans having a precise idea of the fundamentals of these GenAI models and that they should be accompanied by specific actors and objectives. In the situation of *design by inversion*, as in GDAs (Hatchuel et al. 2021), a topology of artifacts is generated, thus implicitly indicating that humans appear as selectors who are able to pick one or several relevant generated designs. The third type of generativity, *design with a fixed language,* appears to put humans at the center of the process, in the sense that, as we have seen, this design exercise is driven by humans themselves with their capacity to combine various pockets of knowledge (Carvajal Pérez et al. 2018), resulting in brand new modeling logic. Finally, the fourth type, *design with no fixed language*, is known to lead to breakthrough innovations when the developed knowledge structure respects the splitting condition (Masson et al. 2016). Rather than recommendations, this raises the question of how humans can augment models by using the latest design theories that are known to foster innovations. In particular, further research could study how to intricate GenAI and C-K theory (Hatchuel and Weil 2008) to make GenAI tend toward a more combinatorial, even expansive, generativity type (Carvajal Pérez et al. 2018). Overall, the issue of human–model interactions remains an avenue for further research to foster the generativity of such GenAI models and make them tend toward *nonfixed-language* generativity. These GenAI models are considered generative tools to be leveraged by humans (Sanders 2000) for increasing generativity. Therefore, this concept opens research avenues regarding how such tools can be used and managed in a generative endeavor or even the extent to which they can change the generative process and the role of designers. Moreover, scientific disciplines, including psycho-cognition and engineering design, have developed methods to foster human generativity (and creativity), which would open research avenues concerning the use of such techniques with GenAI (Rafner et al. 2023). All these opened research avenues could benefit the GenAI community not only by enhancing model generativity but also by producing a bootstrapping effect, resulting in interesting advancements in other scientific disciplines, specifically engineering design (Reich et al. 2008).

### 6.3 A major shift in modeling logic

The logic represented by Lam et al. (2023) for weather prediction is emblematic of a new modeling logic that does not rely on any predicates, unlike more traditional weather prediction techniques. Another famous and emblematic example of this new modelling logic is the identification of Halicin as a major Escherichia Coli inhibitor (Stokes et al. 2020). In this paper, the authors lean on a new representation of molecules as continuous vectors (and not fingerprint vectors as traditionally). Combined with scoring techniques based on neural networks, thus led them to identify Halicin as an E. Coli inhibitor, whereas it was mainly known for its induced mitochondrial dysfunction (Jang et al. 2015). This is a major modeling shift that is also found in GenAI models that generally do not assume any predicates but rather are based only on an unexpected representation of the target artifact. One should insist on this aspect since it is at the heart of the impressiveness of such models. For instance, in the world of GenAI, images are represented as sets of pixels, whereas for humans, images are represented as meanings, feelings or even styles. Similarly, models see a sentence as a set of tokens with a specific order, whereas humans see a sentence as a meaning conveying feelings. One of the major breakthroughs lies in these new accessible representations, which allow for the development of new languages and impressively efficient models.

## 7 Conclusion

What is generative in GenAI? Based on the analysis described above, we are able to answer this question. The latest theories of engineering design, especially C-K theory, provided us with a framework for understanding generativity, giving us with the opportunity to provide a three-step answer.

First, we confirmed that GenAI tools have generative capacities. Second, we showed that the generative mechanisms of the four main families of GenAI models are homogeneous, all of which provide *design by activation* generativity. Third, this characterization provides a theoretical basis for understanding the discussed limits of such models, as well as opportunities for future research, especially those regarding human-GenAI interactions. Moreover, one of the key points of this research is to highlight the existence of a second generative level in GenAI. The level characterized in the previous paragraph concerns the generativity of a model, leading to a new artifact. The second level is the generativity that leads to the design of a new model itself, which enters the *fixed-language* generativity type. However, as impressive as this ability to create new patterns where none existed before is, such a method does not possess strong combinatorial capabilities in the sense that it only represents what is already known and will hardly create new unseen combinations.

Overall, this work shows the limited generative power of GenAI. This has many implications, among which is the notion that such models need to be considered generative tools that must be complemented by humans. This would allow us to leverage the generative power of both humans and GenAI models, which are both based on different representations of the world and thus could open new and rich avenues for undertaking generative endeavors. For the engineering design community, this work is important because it demonstrates the relevance of design theories for understanding and shedding light on contemporary issues and debates around generativity. Moreover, it opens stimulating research avenues toward the intersection with artificial intelligence, which could result in enriching both corpora. To clarify further these research avenues, we mention nascent research on a *design by inversion* framework in the field of artificial intelligence, which could be pursued in the tradition of generative design algorithms. Moreover, the shift in the modeling paradigm mentioned above, which is largely based on access to new representations of the world and the fixed-language generativity of humans when they design models, could indicate the directions of future breakthroughs and the associated new models in domains where digitalization is rolling out. Finally, the engineering design literature indicates that major breakthroughs come from *nonfixed-language* generativity, which is currently accessible only to humans; however, the question of how humans can leverage this *nonfixed-language* generativity to enhance the generativity of GenAI tools is still being raised because of design theories.

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## Declarations

**Conflict of interest** The authors declare no competing interests.

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